

Problem 1

What is the magnitude of the gravitational force between two electrons that are $r = 0.2 \text{ nm}$ apart? The mass of an electron is $m = 9 \times 10^{-31} \text{ kg}$, and the gravitational constant is $\gamma = 6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2$.

Solution

The gravitational force between two particles is given by Newton's law of gravitation:

$$F_g = \gamma \frac{m_1 m_2}{r^2}$$

Since both particles are electrons, we have $m_1 = m_2 = m$:

$$F_g = \gamma \frac{m^2}{r^2}$$

Substitute the given values:

$$F_g = (6.67 \times 10^{-11}) \times \frac{(9 \times 10^{-31})^2}{(0.2 \times 10^{-9})^2}$$

Calculate the force step by step:

$$\begin{aligned} &= (6.67 \times 10^{-11}) \times \frac{81 \times 10^{-62}}{4 \times 10^{-20}} \\ &= (6.67 \times 10^{-11}) \times (20.25 \times 10^{-42}) \\ &= 1.35 \times 10^{-51} \text{ N} \end{aligned}$$

Conclusion: The gravitational force between two electrons at a distance of 0.2 nm is approximately $1.35 \times 10^{-51} \text{ N}$.